

Impact of Slow Data Poisoning Attacks on Smart Traffic Light Control Systems

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Abstract—Smart traffic light control systems increasingly rely on adaptive decision-making based on real-time traffic observations, making them vulnerable to integrity attacks targeting their sensing and control workflows. This paper investigates the impact of a slow data poisoning attack on an adaptive traffic signal controller implemented in a SUMO-based smart traffic simulation environment.

The proposed attack gradually manipulates lane-level traffic features, including queue length, waiting time, and vehicle count, in order to bias the controller's prioritization behavior without directly modifying the physical infrastructure. To ensure statistical validity, the evaluation was conducted over 10 independent simulation runs using randomized seeds.

The results show that the attack significantly distorts the controller's internal behavior, increasing the average green time from 31.86 s to 76.59 s (+140.40%) and the decision score from 11.86 to 181.78 (+1433.07%). In addition, a moderate increase in network-level waiting time (+1.31%) and halting vehicles (+0.82%) was observed.

These findings demonstrate that slow data poisoning attacks can effectively compromise adaptive traffic control logic while remaining relatively stealthy in early stages, leading to gradual performance degradation. This highlights a critical security vulnerability in intelligent transportation systems and emphasizes the need for robust anomaly detection mechanisms.

Index Terms—Smart traffic lights, slow data poisoning attack, intelligent transportation systems, cybersecurity, anomaly detection

I. INTRODUCTION

Intelligent transportation systems (ITS) have become a fundamental component of modern smart cities [3], [7]. Among these systems, smart traffic light control plays a significant role in managing traffic flow, reducing congestion, and improving road safety [3]. These systems rely on real-time traffic data collected from sensors, cameras, and connected devices to dynamically adjust signal timing and optimize traffic movement. In the context of Saudi Arabia, the security and efficiency of these systems are paramount to achieving the goals of Saudi Vision 2030, which prioritizes the development of sustainable, safe, and technologically advanced smart cities.

Despite their advantages, smart traffic light systems are increasingly vulnerable to cybersecurity threats due to their reliance on networked infrastructure and automated decision-making mechanisms [5], [12], [13]. Malicious actors may exploit these systems by manipulating traffic data, disrupting signal coordination, or injecting false information into the control system.

One particularly dangerous type of cyberattack is the slow data poisoning attack. Unlike traditional attacks that cause immediate disruptions, slow poisoning attacks gradually modify system data over time. This gradual manipulation makes the attack difficult to detect because the changes appear subtle and resemble normal traffic variations [1], [8].

Understanding the impact of such attacks is essential for ensuring the security and reliability of intelligent transportation systems. Therefore, this study investigates the effect of slow poisoning attacks on smart traffic light systems and analyzes how gradual data manipulation can influence traffic behavior and system performance.

In the context of adversarial attacks, threat models can be broadly classified into white-box and black-box scenarios. In a white-box attack, the attacker has full knowledge of the system, including its algorithms, internal parameters, and decision rules, allowing precise manipulation of traffic data. In contrast, in a black-box attack, the attacker has limited or no knowledge of the internal structure of the system and instead relies on observing system behavior to gradually influence the input data. Slow data poisoning attacks can be implemented under both models, as the attacker may either directly manipulate traffic signal control parameters in a white-box setting or indirectly influence traffic flow patterns and observed measurements in a black-box setting [8], [9].

The main contributions of this paper can be summarized as follows:

- This study investigates the impact of slow data poisoning attacks on smart traffic light control systems within intelligent transportation systems.
- A realistic traffic simulation environment was developed using the SUMO (Simulation of Urban Mobility) platform and a real-world road network obtained from OpenStreetMap to evaluate the behavior of the traffic system under attack conditions.
- A workflow-level slow poisoning attack model was implemented by gradually manipulating lane-level traffic features, including queue length, waiting time, and vehicle count, in order to bias the controller's adaptive decision-making process.
- Experimental results based on 10 independent simulation runs demonstrate that the proposed attack significantly distorts internal traffic signal control behavior, particularly green-time allocation and controller scoring, while

producing a measurable early-stage impact on traffic performance.

II. BACKGROUND AND RELATED WORK

Cybersecurity threats targeting intelligent transportation systems have attracted increasing research attention in recent years. Several studies have investigated the vulnerability of traffic signal control systems to malicious attacks that manipulate traffic signals or traffic data.

Laszka et al. [3] analyzed the vulnerability of transportation networks to traffic signal tampering attacks and demonstrated that manipulating traffic signal timing can significantly disrupt traffic flow and lead to severe congestion across urban transportation networks.

Other studies have focused on detecting malicious activities in traffic signal infrastructures. Oza et al. [4] proposed a detection mechanism for replay attacks in model-based smart traffic controllers by analyzing inconsistencies between expected and observed signal behavior.

Machine learning techniques have also been explored to improve the detection of cyberattacks in intelligent transportation systems. Chowdhury et al. [1] introduced an intrusion detection framework that analyzes traffic data patterns to identify abnormal traffic behavior caused by malicious interference.

More recently, Li et al. [2] proposed an attack detection framework that combines game theory and generative adversarial networks to enhance the resilience of traffic signal control systems against cyber threats.

Additional studies have explored adversarial attacks in cyber-physical systems and intelligent transportation infrastructures. Recent research highlights the increasing risks associated with data manipulation attacks targeting real-time control systems and emphasizes the need for resilient and secure traffic management solutions [9]–[11], [14], [15].

Although these studies have contributed to improving the security of traffic signal systems, most existing research focuses on sudden or direct attacks that cause immediate disruptions. Limited research has examined slow and stealthy attacks that gradually manipulate traffic data over time. Therefore, this study investigates how slow poisoning attacks can gradually influence traffic signal behavior and traffic congestion within smart traffic light systems.

III. SIMULATION ENVIRONMENT

To evaluate the impact of slow poisoning attacks on smart traffic light systems, a realistic traffic simulation environment was developed using the Simulation of Urban Mobility (SUMO) platform. SUMO is an open-source microscopic traffic simulator widely used in transportation research to analyze vehicle movement and traffic behavior under various conditions [16], [18].

In order to increase the realism of the simulation, a real-world road network was used instead of an artificial map. The road network was obtained from OpenStreetMap (OSM), which provides detailed geographic data of real road infrastructures [17]. The selected map represents a high-density,

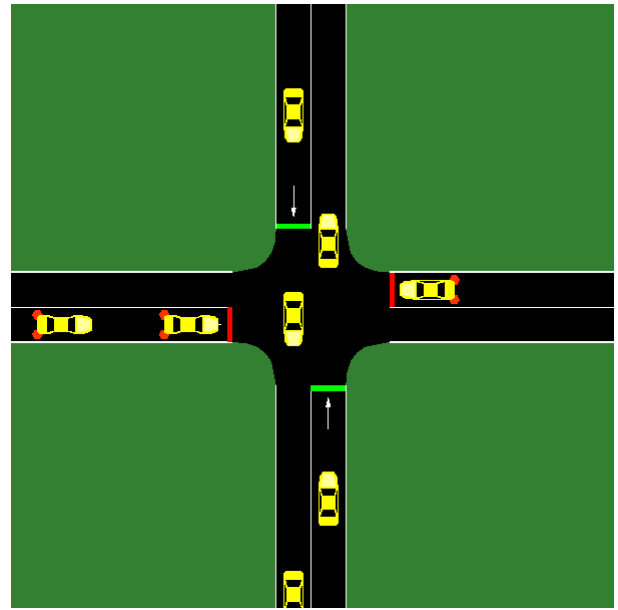


Fig. 1. Traffic simulation environment using SUMO showing the smart traffic light intersection.

four-way urban intersection within a major metropolitan area in Saudi Arabia. The geographic data from OpenStreetMap was then converted into a SUMO-compatible road network using the *netconvert* tool. This conversion process generated a traffic network containing roads, lanes, intersections, and traffic signal configurations.

The simulation focused on a specific traffic intersection where multiple traffic flows converge. Traffic lights were configured within the SUMO environment to regulate the movement of vehicles through the intersection. The traffic signal controller dynamically manages the allocation of green and red signal phases based on the observed traffic conditions.

To simulate realistic traffic behavior, vehicle flows were generated within the network using SUMO traffic demand generation tools. Different vehicle routes were created to represent vehicles entering and leaving the intersection from multiple directions. The simulation monitored key traffic indicators such as vehicle density, queue length, and waiting time at the intersection.

The revised experimental setup used a simulation horizon of 1000 steps. Vehicle flows were active during the main traffic generation interval, and the controller evaluated traffic conditions every 10 simulation steps. To improve experimental validity, 10 independent runs were conducted using randomized simulation seeds. This enabled statistical comparison between the baseline and attack scenarios using mean and standard deviation.

The simulation environment used in this study is illustrated in Fig. 1.

IV. ATTACK MODEL

In this study, a slow data poisoning attack is introduced to analyze its impact on smart traffic light control systems. Data

poisoning attacks belong to a class of adversarial attacks in which an attacker intentionally injects manipulated data into a system in order to influence its decision-making process [1], [8]. Unlike traditional cyberattacks that cause immediate disruption, slow poisoning attacks introduce small and incremental modifications to the system data over time, making the attack significantly more difficult to detect during the early stages.

The main characteristic of a slow poisoning attack is that the manipulated data remains very close to normal traffic patterns. Instead of introducing large abnormal values, the attacker injects subtle incremental changes that gradually accumulate over time. Because these modifications resemble natural variations in traffic conditions, the attack can remain stealthy while progressively influencing the traffic signal controller's decision-making process [2], [14].

To implement the slow poisoning attack within the traffic control system, a gradual manipulation was applied to the lane-level traffic observations used by the controller rather than directly modifying the signal timing parameters. Specifically, the attack incrementally increased the perceived queue length, waiting time, and vehicle count of a selected target lane.

The objective of this manipulation was to bias the controller's decision-making process by making the targeted lane appear more congested than it actually is, without triggering immediate abnormal system behavior. Because each modification was small and introduced gradually, the changes remained consistent with normal traffic fluctuations.

Over multiple control intervals, the accumulated perturbations caused the controller to repeatedly prioritize the attacked lane, leading to an imbalance in green time allocation. As a result, the traffic signal behavior became progressively less efficient, contributing to increased waiting times and localized congestion at the intersection.

The implementation of the slow poisoning attack within the SUMO simulation environment is illustrated in Fig. 2. In this experiment, no additional vehicles were physically injected into the traffic network. Instead, the attack operates at the data level by manipulating the traffic observations used by the controller.

Specifically, the perceived values of queue length, waiting time, and vehicle count for a selected target lane are gradually increased over time. This manipulation creates the illusion of congestion without altering the actual vehicle flow in the simulation.

As a result, the controller is misled into prioritizing the attacked lane and assigning longer green times, leading to imbalanced traffic signal behavior. The visualization in Fig. 2 demonstrates how the attack influences the control decisions through falsified traffic data rather than direct physical changes in the network.

The attack scenario was implemented within the SUMO (Simulation of Urban Mobility) traffic simulation platform. SUMO is an open-source microscopic traffic simulator widely used in transportation research to analyze vehicle movement and traffic control strategies [3], [16]. In this study, a real-

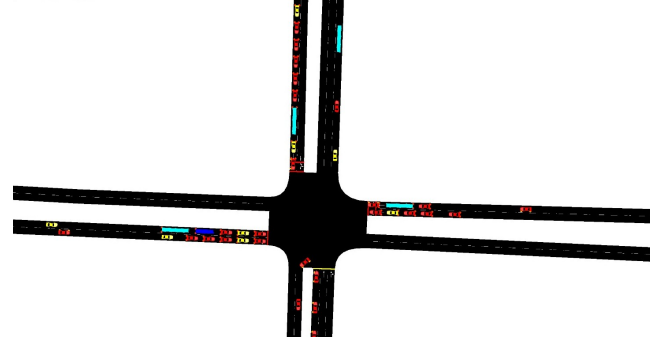


Fig. 2. SUMO simulation showing the attack scenario at the selected intersection. The attack targets the controller by poisoning lane-level traffic observations, leading to biased signal timing decisions.

world road network obtained from OpenStreetMap (OSM) [17] was used to create a realistic traffic environment representing a typical high-volume traffic intersection in a major Saudi Arabian city. The geographic data from OpenStreetMap was converted into a SUMO-compatible traffic network using the *netconvert* tool.

Vehicle flows were generated within the simulation to represent realistic traffic conditions at the intersection. The slow poisoning attack was then introduced by gradually manipulating the lane-level traffic observations used by the controller rather than directly modifying signal timing parameters. The experiment was conducted on a personal computer equipped with an Intel Core processor, 16 GB of RAM, and the Windows operating system. The SUMO graphical interface and simulation tools were used to monitor traffic behavior and evaluate the impact of the attack on traffic congestion and vehicle waiting times.

To provide a more formal representation of the slow poisoning attack, the traffic signal timing under attack can be modeled using a vector-based formulation:

$$\mathbf{T}_{attack} = \mathbf{T}_{normal} + \Delta(t) \quad (1)$$

where $\mathbf{T}_{normal}$ represents the normal signal timing vector and $\Delta(t)$ represents a time-dependent perturbation vector introduced by the attacker.

A. Mathematical Model of Slow Poisoning Attack

The normal traffic signal timing vector is defined as:

$$\mathbf{T}^{(0)} = \begin{bmatrix} T_1^{(0)} \\ T_2^{(0)} \\ \vdots \\ T_n^{(0)} \end{bmatrix} \quad (2)$$

At time step t , the attacked signal timing becomes:

$$\mathbf{T}(t) = \mathbf{T}^{(0)} + \Delta(t) \quad (3)$$

The perturbation accumulates over time:

$$\Delta(t) = \sum_{k=1}^t \mathbf{u}(k) \quad (4)$$

where

$$\mathbf{u}(k) = \begin{bmatrix} u_1(k) \\ u_2(k) \\ \vdots \\ u_n(k) \end{bmatrix} \quad (5)$$

and each perturbation component is defined as:

$$u_i(k) = \alpha_i + \beta_i \ln(1+k) + \gamma_i \varepsilon_i(k) \quad (6)$$

where α_i represents a constant bias injected by the attacker, β_i is a logarithmic growth factor that controls the rate of timing degradation to ensure stealthiness, and γ_i is the noise coefficient associated with the stochastic term $\varepsilon_i(k)$, used to mimic natural traffic fluctuations and evade detection.

In matrix form:

$$\mathbf{u}(k) = \mathbf{A}\mathbf{1} + \mathbf{B}\mathbf{1} \ln(1+k) + \mathbf{\Gamma}\varepsilon(k) \quad (7)$$

$$\mathbf{A} = \begin{bmatrix} \alpha_1 & 0 & \cdots & 0 \\ 0 & \alpha_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \alpha_n \end{bmatrix} \quad (8)$$

$$\mathbf{B} = \begin{bmatrix} \beta_1 & 0 & \cdots & 0 \\ 0 & \beta_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \beta_n \end{bmatrix} \quad (9)$$

$$\mathbf{\Gamma} = \begin{bmatrix} \gamma_1 & 0 & \cdots & 0 \\ 0 & \gamma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \gamma_n \end{bmatrix} \quad (10)$$

The final accumulated perturbation model is:

$$\mathbf{T}(t) = \mathbf{T}^{(0)} + \sum_{k=1}^t (\mathbf{A}\mathbf{1} + \mathbf{B}\mathbf{1} \ln(1+k) + \mathbf{\Gamma}\varepsilon(k)) \quad (11)$$

$$\|\mathbf{u}(k)\|_2 \leq \theta_s \quad (12)$$

where θ_s is a stealthiness threshold that constrains the perturbation magnitude so that the attack remains close to normal traffic behavior.

In the practical implementation used in this study, the attack is realized at the workflow level by gradually manipulating lane-level traffic observations, including queue length, waiting time, and vehicle count, which in turn bias the controller toward assigning longer green durations. Thus, the above mathematical model provides a formal representation of gradual perturbation behavior, while the simulation implementation

applies this concept through falsified traffic observations rather than direct modification of the physical signal infrastructure.

For clarity, the observed traffic feature vector for a controlled lane at time step t can be written as:

$$\mathbf{X}(t) = [q(t), w(t), v(t)] \quad (13)$$

where $q(t)$ is queue length, $w(t)$ is waiting time, and $v(t)$ is vehicle count.

Under the slow poisoning attack, the attacked lane is represented as:

$$\mathbf{X}'(t) = \mathbf{X}(t) + \Delta_x(t) \quad (14)$$

where $\Delta_x(t)$ is a gradually increasing perturbation applied to the observed traffic features over time. The poisoned observations are then used by the controller to compute lane priority scores and assign adaptive green-time durations.

V. RESULTS AND DISCUSSION

The simulation results illustrate the impact of the slow poisoning attack on the performance of the smart traffic light control system. To evaluate the effect of the attack, multiple traffic performance metrics were analyzed, including waiting time, halting vehicles, and traffic signal behavior (green-time allocation). These metrics provide insight into how gradual manipulation of traffic observations influences both controller decisions and overall traffic performance over time.

Figure 4 presents a time-series comparison between the baseline scenario and the attack scenario across the simulation steps. Under normal operation (baseline), the controller maintains relatively stable green-time allocation, resulting in balanced traffic flow and moderate variations in waiting time and halting vehicles. In contrast, under the attack scenario, the controller behavior is progressively distorted due to the injected perturbations in the observed traffic data.

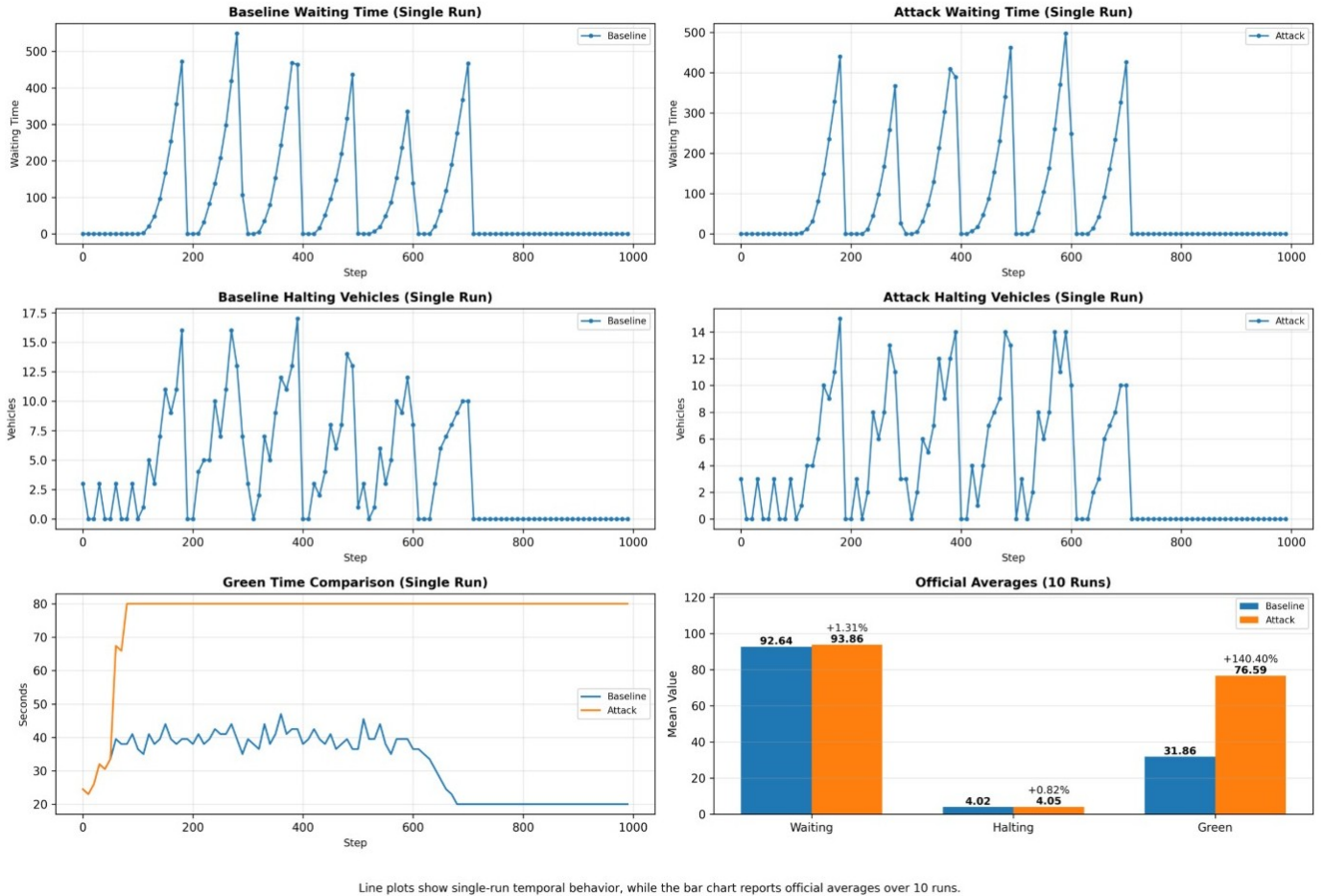
Specifically, the figure shows that the most pronounced impact of the attack occurs in the green-time allocation, where the assigned green duration increases significantly and stabilizes at a higher level compared to the baseline. This indicates that the controller is being biased toward a specific lane or traffic phase.

At the traffic level, the impact is more subtle but still observable. The waiting time exhibits slightly higher peaks under the attack scenario, while the number of halting vehicles shows increased fluctuations, indicating less efficient traffic flow. However, these changes remain relatively moderate, reflecting the stealthy nature of the slow poisoning attack, which gradually alters system behavior without immediately causing severe congestion.

Overall, the figure demonstrates that slow poisoning attacks primarily affect the internal decision-making logic of the traffic controller, with gradual and delayed effects on observable traffic conditions.

These observations are further supported by the enhanced visualization, which combines single-run temporal behavior with averaged results.

Baseline vs Attack: Single-Run Temporal Behavior and Official Multi-Run Averages



Line plots show single-run temporal behavior, while the bar chart reports official averages over 10 runs.

Fig. 3. Enhanced comparison between baseline and attack scenarios. The line plots show single-run temporal behavior, while the bar chart presents the official average results over 10 independent simulation runs.

The enhanced visualization in Fig. 3 provides a comprehensive comparison between the baseline and attack scenarios by combining both temporal and statistical perspectives. The line plots illustrate the single-run temporal behavior of key traffic performance metrics, including waiting time, halting vehicles, and green-time allocation, allowing observation of how the attack evolves over simulation steps.

It can be observed that the most significant deviation occurs in the green-time allocation, where the attack scenario rapidly stabilizes at a higher value compared to the baseline, indicating a strong bias in the controller's decision-making process. In contrast, traffic-level metrics such as waiting time and halting vehicles exhibit more subtle differences, reflecting the stealthy nature of the slow poisoning attack.

To further validate these observations, the bar chart presents the official average results computed over 10 independent simulation runs. The statistical results confirm that the attack leads to a substantial increase in average green time (from 31.86 s to 76.59 s, +140.40%), while only moderate increases are observed in waiting time (+1.31%) and halting vehicles

(+0.82%).

These findings highlight that the slow poisoning attack primarily targets the internal control logic of the traffic signal system before causing noticeable degradation in observable traffic performance. This behavior demonstrates the stealthiness and effectiveness of gradual data manipulation attacks in intelligent transportation systems.

Furthermore, the integration of both temporal and aggregated views allows for a more comprehensive interpretation of the attack dynamics, where short-term behavioral patterns align with long-term statistical trends. This consistency reinforces the reliability of the experimental findings and confirms that the observed deviations are not isolated artifacts but systematic effects induced by the attack.

Overall, the results emphasize that slow poisoning attacks can remain inconspicuous during early stages while progressively influencing controller behavior, highlighting the need for advanced monitoring mechanisms capable of detecting gradual anomalies in traffic signal systems.

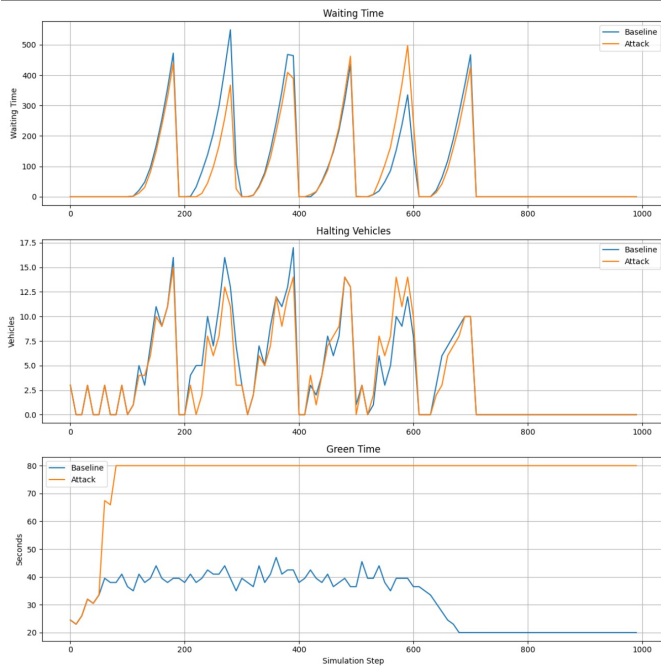


Fig. 4. Combined time-series comparison between baseline and attack scenarios, showing the evolution of waiting time, halting vehicles, and green time over the simulation steps.

The results indicate that the slow poisoning attack primarily affects the controller-level behavior rather than causing immediate severe degradation in overall traffic performance.

As shown in Fig. 4, the most significant impact of the attack is observed in the green time allocation, where the controller increasingly favors the attacked lane. This leads to a clear imbalance in signal timing decisions over time.

In contrast, traffic-level metrics, including waiting time and halting vehicles, exhibit only slight increases under the attack scenario. This suggests that the attack initially manifests as a manipulation of control logic before translating into more noticeable traffic congestion.

TABLE I

STATISTICAL COMPARISON OF BASELINE AND ATTACK SCENARIOS OVER 10 INDEPENDENT SIMULATION RUNS

Metric	Baseline (Mean $\pm$ Std)	Attack (Mean $\pm$ Std)	Change (%)
Average Waiting Time (s)	92.64 $\pm$ 6.52	93.86 $\pm$ 6.58	+1.31
Average Halting Vehicles	4.02 $\pm$ 0.16	4.05 $\pm$ 0.16	+0.82
Average Green Time (s)	31.86 $\pm$ 0.17	76.59 $\pm$ 0.02	+140.40
Average Queue Length	0.20 $\pm$ 0.00	0.20 $\pm$ 0.00	0.00
Average Decision Score	11.86 $\pm$ 0.17	181.78 $\pm$ 0.15	+1433.07
Average Network Vehicles	52.36 $\pm$ 0.44	52.52 $\pm$ 0.29	+0.31

Table I presents a statistical comparison between the baseline and attack scenarios based on 10 independent simulation runs. The reported values correspond to the mean and standard deviation of each metric.

The attack has its strongest impact on controller-level behavior. In particular, the average green time increased from 31.86 s (± 0.17) to 76.59 s (± 0.02), representing a 140.40% increase. Similarly, the average decision score increased from 11.86

(± 0.17) to 181.78 (± 0.15), corresponding to a 1433.07% increase. These results confirm that the attack effectively biases the controller's internal prioritization mechanism.

At the traffic level, the impact remains moderate but measurable. The average waiting time increased from 92.64 s (± 6.52) to 93.86 s (± 6.58), while the number of halting vehicles increased from 4.02 (± 0.16) to 4.05 (± 0.16). This indicates that the attack begins to affect traffic performance without immediately causing severe congestion.

Overall, these findings demonstrate that slow data poisoning attacks can significantly manipulate adaptive traffic signal behavior while remaining relatively stealthy in early stages, causing gradual performance degradation before large-scale congestion becomes evident.

VI. CONCLUSION

This study investigated the impact of slow data poisoning attacks on adaptive smart traffic light control systems within intelligent transportation environments. Using a SUMO-based simulation with a road network derived from the Jeddah region, we demonstrated that gradual manipulation of lane-level traffic observations can significantly distort the internal behavior of a traffic signal controller.

The experimental results, based on 10 independent simulation runs, show that the attack primarily affects controller-level behavior, particularly in green-time allocation and decision scoring, while producing a measurable but moderate impact on traffic-level performance. These findings indicate that slow poisoning attacks can compromise adaptive traffic control logic before severe congestion becomes fully visible.

Overall, this work highlights the need for trend-based anomaly detection and resilient traffic control mechanisms capable of identifying gradual integrity attacks in intelligent transportation systems. Future work will focus on evaluating the attack under denser traffic conditions, multi-intersection environments, and exploring learning-based detection approaches such as GAN-based anomaly detection.

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